Automation in Construction 106 (2019) 102894

Contents lists available at ScienceDirect

Automation in Construction

journal homepage: www.elsevier.com/locate/autcon

Automatic detection of hardhats worn by construction personnel: A deep learning approach and benchmark dataset

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ARTICLE INFO

Keywords:
Construction sites safety
Hardhat wearing detection
Computer vision
Convolutional neural network
Reverse progressive attention

ABSTRACT

Hardhats play an essential role in protecting construction individuals from accidents. However, wearing hardhats is not strictly enforced among workers due to all kinds of reasons. To enhance construction sites safety, the majority of existing works monitor the presence and proper use of hardhats through multi-stage data processing, which come with limitations on adaption and generalizability. In this paper, a one-stage system based on convolutional neural network is proposed to automatically monitor whether construction personnel are wearing hardhats and identify the corresponding colors. To facilitate the study, this work constructs a new and publicly available hardhat wearing detection benchmark dataset, which consists of 3174 images covering various on-site conditions. Then, features from different layers with different scales are fused discriminately by the proposed reverse progressive attention to generate a new feature pyramid, which will be fed into the Single Shot Multibox Detector (SSD) to predict the final detection results. The proposed system is trained by an end-to-end scheme. The experimental results demonstrate that the proposed system is effective under all kinds of on-site conditions, which can achieve 83.89% mAP (mean average precision) with the input size 512 × 512.

1. Introduction

While other industries have undergone profound changes in productivity via automation and artificial intelligence, the business of building has not. Construction is still one of the least digitized industries worldwide and incurs a huge demand for workers. However, construction is one of the most dangerous job sectors. According to the Bureau of Labor Statistics (BLS) in the United States, the fatal injury rate for the construction industry is higher than the national average in this category for all industries [1]. In China, over 2850 construction workers died from building construction activities during 2012–2016 with an average of 1.57 fatalities each day [2]. The construction fatalities are always caused by the combination of different factors. Among them, traumatic brain injuries account for about 24% of all construction fatalities in the United States [3,4]. Therefore, there is a critical demand for on-site safety policies and procedures to enhance construction sites safety. The use of personal protective equipment (PPE), such as a hardhat, provides effective risk management. Nonetheless, the workers do not exactly follow the construction sites safety regulations due to all kinds of reasons, even if they have been previously educated and trained [5]. Consequently, existing safety inspection practices are predominantly reliant on inspectors' manual monitoring and reporting.

To facilitate the safety monitoring work of construction sites safety inspectors, a considerable amount of studies have been published for automatic hardhat wearing detection [6–19]. Among them, the vision-based techniques [6–8,10–14,17–19] occupy a dominant position compared with the high-cost sensor-based solutions [9,15,16]. Conventionally, given a frame from the surveillance camera on construction sites, the vision-based techniques perform hardhat wearing detection through multiple stages, which mainly consist of pedestrians detection [20,21], hardhats localization and hardhats identification. Although extensive researches have been carried out, the vision-based hardhat wearing detection remains challenging. First, large variability in the background and pedestrian states are caused by various on-site conditions so that researches under specific scenes have difficulties in extending them to other construction sites. Meanwhile, the presence of small-scale individuals that are relatively far from the camera makes it hard to distinguish them from the background clutters and other overlapped instances. In addition, several individuals may be present in the same image region, partially occluding each other. This also makes hardhat wearing detection difficult. Finally, until now there are no publicly-available open datasets for developing and evaluating hardhat wearing detection algorithms under various circumstances.

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https://doi.org/10.1016/j.autcon.2019.102894
Received 27 February 2019; Received in revised form 23 June 2019; Accepted 26 June 2019
Available online 16 July 2019
0926-5805/ © 2019 Published by Elsevier B.V.

In this paper, we specifically address the task of hardhat wearing detection on construction sites. The goal is to identify whether all individuals on construction sites are wearing hardhats and, if so, which colors the hardhats are. The difficulty of this task mentioned above makes it problematic to rely on any type of multi-stage approaches with hand-craft features. To solve this problem, the convolutional neural network (CNN) is introduced here, which is capable of automatic features learning and delivers superior performance in the field of computer vision in comparison with approaches based on traditional image processing. Our work is motivated by the development of the Single Shot Multibox Detector (SSD) [22], which detects objects by directly regressing bounding boxes via a single CNN. However, the SSD commonly fails to detect small-scale objects because of its inherent properties of weak features at the bottom layers of high resolution, even if it has built a pyramidal feature hierarchy. In order to detect small-scale hardhats effectively, we develop a novel aggregation framework combined with the presented reverse progressive attention (RPA), which propagates the semantically strong features back to the bottom layers progressively. In this way, the features from bottom layers, which may correspond to objects with smaller scales, can effectively be learned to contribute to the detection results.

To summarize, this paper contains the following contributions:

1) In contrast to commonly used multi-stage approaches, this study provides a one-stage solution to automatically identifying hardhats wearing on construction sites. The proposed system is end-to-end trainable. Also, it is more effective and robust with scale variation, viewpoint changes and occlusions in congested scenes.
2) We propose to aggregate multiscale convolutional features discriminately with reverse progressive attention. Such a scheme demonstrates its superiority in small-scale hardhats detection on various construction sites.
3) We have constructed a new hardhat wearing detection benchmark. It provides 3174 images containing 18,893 hardhats instances. These images cover large variations in scenes and illuminance, also including examples with occlusion. Each instance in the benchmark is annotated with a class label and its bounding box. We call this benchmark GDUT-Hardhat Wearing Detection (GDUT-HWD).

2. Related works

Conventionally, the methods of hardhat wearing detection can be divided into two categories: sensor-based detection and vision-based detection.

Sensor-based detection methods [9,15,16] have focused on remote locating and tracking techniques, such as radio frequency identification (RFID) and wireless local areas networks (WLANs). Kelm et al. [15] designed a mobile RFID portal for checking PPE compliance of personnel. However, the RFID readers located at the construction sites entrance are incapable of checking non-entrance areas. Besides, it is unable to identify whether the PPE is being worn. Dong et al. [16] developed the real-time location system (RTLS) and virtual construction for a worker's location tracking to decide whether the worker should wear a hardhat and to transmit a warning. To identify whether a hardhat was being worn, a pressure sensor was placed in the hardhat, and then the pressure information was transmitted via Bluetooth for monitoring. Zhang et al. [9] developed a smart hardhat system using an Internet of Things (IoT)-based architecture. To determine the status of hardhat-use, an infrared beam detector and a thermal infrared sensor were located inside the hardhat. The worker's hardhat-use was confirmed when both infrared beam detector and thermal infrared sensor were activated. Generally, existing sensor-based methods relying on physical tags or sensors employed in PPE have difficulties in identifying whether any individuals on the construction sites are wearing hardhats or not. Besides, the practical use of the tags or sensors will lead to significant costs with massive productions.

Compared with the sensor-based detection methods, vision-based techniques have been received increasing attention. In this regard, the use of regular cameras and the advancements in computer vision and pattern recognition techniques have created solid foundations for vision-based hardhat wearing detection. Wu et al. [19] proposed a color-based hybrid descriptor comprised of local binary patterns (LBP), hu moment invariants (HMI), and color histograms (CH) to extract features of hardhats with different colors. Then a hierarchical support vector machine (H-SVM) was used to classify the features into four classes (red, yellow, blue, and non-hardhat). Li et al. [8] developed an approach to detect workers based on the ViBe algorithm and the C4 human classification framework. To perform hardhat wearing detection, a prior knowledge-based head region was cropped and the HSV color space was utilized to carry out the classification. Mneymneh et al. [10] proposed a framework to monitor hardhat wearing by first detecting moving objects using standard deviation matrix (SDM) and then classifying human using the aggregate channel feature-based object detector [23]. Afterwards, a cascade object detector based on histogram of the oriented gradients (HOG) features searched for hardhats in the upper region of the identified personnel, which would be fed into the color-based classification component In general, these multi-stage methods rely heavily on hand-crafted features to detect individuals on construction sites. Consequently, they may fail in the cases of complicated scenes with weather variability, different viewpoints and occlusions.

Recently, the growing popularity of deep learning-based object detection has motivated CNN-based hardhat wearing detection. Fang et al. [17] proposed a Faster R-CNN [24] based method to automatically detect construction workers' non-hardhat-use (NHU). A total of 81,000 image frames were collected from various construction sites as a training dataset to train the Faster R-CNN model. In the training phase, the worker-of-interest (WOI) in the image was annotated as the ground truth for training. In the test phase, the NHU workers will be detected and the rest will be considered the background. Identifying hardhat colors is unfeasible in this work. Hardhat colors can signify different roles on construction sites, which provide an accessible way to enhance construction safety management. Besides, the Faster R-CNN model heavily relying on the information extracted by the upper features do not make full use of the bottom details, according to [24], which may compromise detection performance on the NHU workers at different scales in the images.

This paper presents a new automatic hardhat wearing detection algorithm based on the SSD framework to address the limitations of the previous works. We demonstrate the superiority of the approach in our released benchmark dataset, which has the benefits of developing new remarkable algorithms for the research community and industrial applications.

3. Methodologies

Images collected from the cameras placed on the construction sites may involve a number of challenging issues such as scale variations, perspective distortion and partial occlusions. Existing multi-stage hardhat wearing detection methods usually fail to effectively generalize to various on-site scenes because of the hand-crafted features designed to deal with specific cases. To solve the problem, in this paper, we propose a one-stage and data-driven CNN method based on the SSD framework. An overview of our architecture is shown in Fig. 1.

3.1. Proposed hardhat wearing detection framework

The SSD is a general object detector built on top of a “backbone” network that ends with some convolutional layers. The insightful idea of SSD lies in using the intermediate feature maps in different layers of different scales, which detects relatively small objects in higher resolution feature maps and big objects in lower resolution feature maps.

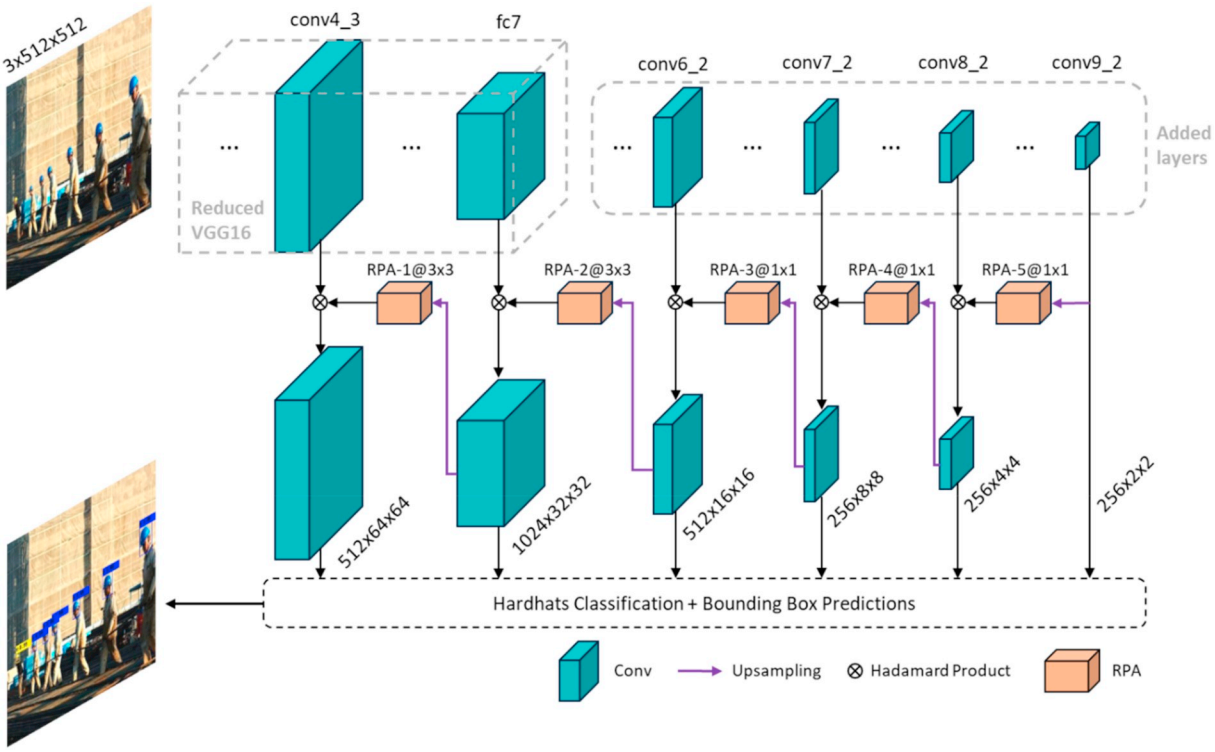


Fig. 1. Architecture of one-stage hardhat wearing detection model. A reduced VGG16 network [25] is adopted here. The feature pyramid fed into the multibox is generated by RPA module of different stages. For each RPA module, the Conv layers use a different kernel size according to the receptive field of different stages. The final result is produced through hardhats classification and bounding boxes regression. More detection details can be found in [22].

Mathematically, it can be defined as follows

$$\Theta_l = \mathscr{F}_l(\Theta_{l-1}) = \mathscr{F}_l(\mathscr{F}_{l-1}(\cdots\mathscr{F}_1(\mathbf{X}))) \tag{1}$$

$$Detections = T(\tau_l(\Theta_l),\cdots,\tau_{l-n}(\Theta_{l-n})),\ l > n > 0 \tag{2}$$

where Θ_l is the feature maps in the l-th layer. $\mathscr{F}_l$ denotes the non-linear mapping to transform the feature maps in the $(l-1)$-th layer to the l-th layer, which could be the composition of the convolutional layers, ReLU layers, and pooling layers et al. $\mathscr{F}_1(\mathbf{X})$ is the first nonlinear block to transfer the input image $\mathbf{X}$ to the feature maps of the first layer. Additionally, $\tau_l(\cdot)$ converts the feature maps of the l-th layer to the detection results for a specific scale. Finally, T combines all n intermediate results and achieves the final detection.

According to Eq. (2), the SSD performing well relies on a strong assumption that every Θ_l has to be sophisticated enough. By sophistication, it means that Θ_l has enough resolution to represent the fine details of the specific-scale objects and contains high-level context information of the objects. However, Θ_l is much deeper than Θ_{l-n} when n is large, which makes $\tau_{l-n}(\cdot)$ commonly fail to detect small-scale objects, because the features in Θ_{l-n} of high resolution are weakly predictive.

We propose a reverse progressive attention driven framework which intelligently selects features for aggregation. Powerful attentive features are generated to conduct hardhats bounding boxes prediction. Compared with the original SSD, the proposed framework can be defined as follows.

$$\begin{aligned}\Phi_l &= \mathscr{D}_l(\Theta_l,\Theta_{l-1})\Phi_{l-1} = \mathscr{D}_{l-1}(\Theta_{l-1},\Phi_l)\vdots\Phi_{l-n} \\ &= \mathscr{D}_{l-n}(\Theta_{l-n},\Phi_{l-n+1})Detections = \mathrm{T}(\tau_l(\Phi_l),\cdots,\tau_{l-n}(\Phi_{l-n})),\ l > n > 0\end{aligned} \tag{3}$$

where $\mathscr{D}_l$ is the proposed reverse progressive attention function in which top contributing feature maps are considered and outputs a new feature representation Φ_l of the same dimensionality to Θ_l. Unlike Θ_l in Eq. (1), Φ_l now not only shares the same resolution as Θ_l, but also incorporates the features extracted in the deeper layers.

3.2. Reverse progressive attention

For the small-scale hardhat wearing detection, due to repeated down-sampling operations such as pooling and convolution, the top layers of small resolution contain fewer features for detection. In contrast, the bottom layers of large resolution keep rich spatial details but include semantically weak features. To accurately locate the hardhats regions and identify the colors of the hardhats simultaneously, it is necessary to combine multi-level features together. However, Existing methods concatenating multi-level features indiscriminately are defective owing to the redundant details and distraction from background. Attention mechanism learning to weight the multi-scale features at each pixel location has become increasingly prevalent in salient object detection [26] and semantic segmentation [27]. Inspired by the attention mechanism, we propose the reverse progressive attention (RPA), which progressively encodes multi-level contextual information to generate more abstract features for hardhat wearing detection. Given the features $\Theta_{l-1} \in \mathbb{R}^{C/2\times H\times W}$ in the $(l-1)$-th layer, the spatial attention maps α_{l-1} is obtained by

$$\Psi_{l-1} = \mathbf{W_s} * UP(\Theta_{l-1})_2 + \mathbf{b_s} \tag{4}$$

$$\alpha_{l-1} = \mathrm{Sigmoid}(\mathbf{W_m} * \Psi_{l-1} + \mathbf{b_m}) \tag{5}$$

where $*$ denotes convolution operation, $\mathbf{W_s} \in \mathbb{R}^{C/2\times 3\times 3}$ and $\mathbf{W_m} \in \mathbb{R}^{1\times k\times k}$ represent convolution filters. $\mathbf{b_s} \in \mathbb{R}^{C/2}$ and $\mathbf{b_m} \in \mathbb{R}^1$ are the bias parameters. $UP(\cdot)_2$ denotes upsampling feature maps by a factor of 2. Now the spatial attention weights $\alpha_{l-1} \in [0,1]^{1\times 2H\times 2W}$ of the semantically strong features Θ_{l-1} have been generated, which serve as a guidance for Θ_l to selectively produce new attentive features, as calculated in Eq. (6).

$$\Phi_l^i = \alpha_{l-1} \circ \Theta_l^i \tag{6}$$

where $i \in \{1,\ldots,\widetilde{C}\}$, Φ_l^i, $\Theta_l^i \in \mathbb{R}^{1\times 2H\times 2W}$ and $\circ$ denotes Hadamard

product operation. In our proposed aggregation framework, the attentive features Φ_l being exploited to detect the final results are produced stage by stage. Meanwhile, the attention map α_{l-1} is learned in an unsupervised manner without any segmentation annotations.

4. Experimental results

Our experiments are all based on VGG16, which is pre-trained on the ILSVRC CLS-LOC dataset [28]. We fine-tune a reduced version of VGG16 using Stochastic Gradient Descent (SGD) with initial learning rate 10^{-3}, 0.9 momentum, 0.0005 decay, and batch size 2. All experiments are conducted on a single machine with a Nvidia Quadro M2000M 4GB GPU. The data and the CNN models built on Caffe [29] are publicly available.[1]

4.1. Dataset statistics

The proposed benchmark dataset GDUT-HWD has been generated by downloading Internet images retrieved by search engines using keywords, which covers large variations in scenes, visual ranges, illuminations, individual postures and occlusions. The number of images of GDUT-HWD is 3174, which has been split into 1587 for training (GDUT-HWD trainval) and 1587 for testing (GDUT-HWD test). It contains 18,893 instances falling into 5 classes and each instance is annotated with a class label and its bounding box. It is notable that the number of small-scale instances (area $\leq 32^2$ pixels) is largest in this dataset, which brings challenges to hardhat wearing detection. Therefore, to test the ability of a detector over different sizes of the objects, we also divide the instances into three scale categories according to their sizes: small objects (area $\leq 32^2$ pixels), medium objects ($32^2 <$ area $\leq 96^2$) and large objects (area $> 96^2$ pixels). Instances per class are shown in Table 1 and the number of instances in each scale category is shown in Fig. 2.

4.2. Evaluation metrics

The widely used mean average precision (mAP) and the per-class average precision (AP) are adopted to evaluate the performance of different models. The AP is to calculate the area under the curve of the *Precision* × *Recall* curve while the mAP is the AP averaged over 5 classes. The precision and recall are calculated for each class and are defined as follows

$$\text{Precision} = TP/(TP + FP) \tag{7}$$

$$\text{Recall} = TP/(TP + FN) \tag{8}$$

where *TP* is defined as the number of the correct detections with an Interaction of Union (IoU) ≥ 0.5. *FP* is the number of the wrong detections, while *FN* is the number of the ground truth not detected.

4.3. Results and comparisons

Previous studies have identified the effectiveness of scene-specific non-hardhat-use detection, but until now no publicly-available open datasets have been released to be conducted for performance comparisons of existing methods. Therefore, we create two baselines for our released GDUT-HWD, which include mAP averaged over 5 classes and AP for object scale. Meanwhile, we have released the models primarily deployed in a system of low memory and real-time processing. All results are based on CNN-based object detectors and are shown in Tables 2 and 3.

[1] https://github.com/wujixiu/helmet-detection/tree/master/hardhat-wearing-detection.

Table 1
The number of instances in the GDUT-HWD dataset. Note that the label "none" indicates the person without wearing a hardhat.

Label	Numbers		
	Trainval	Test	Total
Blue	1251	1361	2612
White	1813	1968	3781
Yellow	1936	1962	3898
Red	2148	2083	4231
None	2171	2200	4371
Small	4237	4713	8950
Medium	4098	3826	7924
Large	984	1035	2019

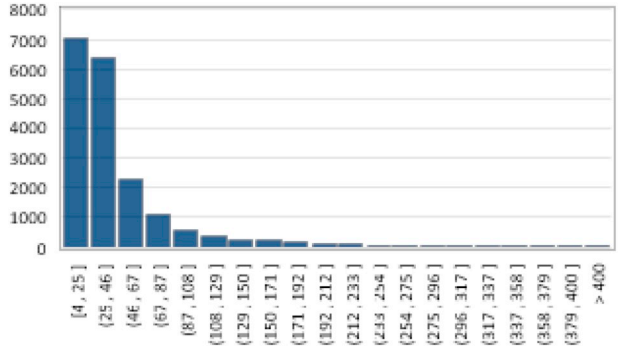


Fig. 2. The number of instances in each scale category.

4.3.1. Benchmark results on GDUT-HWD

We report the detection results on GDUT-HWD test set in Table 2, in which upper results are from state-of-the-art full-sized detection models, while the lower part results are from our proposed models. As shown in Table 2, our low-resolution SSD-RPA300 model is already more accurate, surpassing Faster R-CNN by 9.55% mAP and SSD300 by 2.12% mAP. If we take our model trained on a larger 512 × 512 input image, the best result of 83.89% mAP is achieved, which is slightly better than SSD512 by 0.62% (83.89% vs 83.27%). The results demonstrate that the sizes of input images are critical for the detection performance of the SSD-based models since objects in the images can still remain strong spatial information in some of the deep layers. It is also indicated in Table 2 that the detection accuracy significantly increases when aggregating multi-scale feature maps for prediction. For example, the Faster R-CNN-based model adopted by Fang et al. has the worst performance for its limitation of transforming a single-scale feature map *conv5* for detection.

Additionally, the results also show the superiority of the proposed model to assist in relatively small hardhat wearing detection. As shown in Table 2, even though SSD-RPA300 and SSD-RPA512 have not shown the advantages for medium and large instances, it should be noted that the SSD-RPA300 model outperforms the other methods for small hardhats when we adopt the same input resolution, which is better than ResNet-50 [30] + FPN [31] by 1.66% (52.09% vs 50.43%). When inputting images with increasing sizes, our SSD-RPA512 model still achieves the highest mAP (67.05%) for small instances, which is higher than the SSD512 model by 0.92%. Although the SSD-RPA512 does not show significant improvement for small instances with higher image sizes and the performance drops slightly in detecting the medium and large instances, it confirms that RPA help boosts the performance of the hardhats wearing detectors and the small instances detection, especially with smaller image sizes. The performance drop in detecting the medium and large instances may ascribe to emphasizing the features extraction at small scales excessively, which make the features of top layers less discriminative for medium and large instances. Fig. 3 shows

Table 2
Detection results with respect to GDUT-HWD test. An Interaction of Union (IoU) threshold of 0.5 is adopted here.

Method	Input size	Backbone	mAP	Blue	White	Yellow	Red	None	Avg. precision, area		
									Small	Medium	Large
Faster R-CNN [17]	300 × 500	VGG16	65.67	70.80	68.03	69.57	60.95	59.01	37.86	80.75	86.67
ResNet + FPN [31]	384 × 640	ResNet-50	73.26	77.61	75.49	76.86	69.67	66.66	50.43	84.83	90.32
SSD300 [22]	300 × 300	VGG16	72.64	77.43	76.32	77.83	67.96	63.65	48.25	81.69	89.05
SSD512 [22]	512 × 512	VGG16	83.27	86.15	85.46	88.16	**80.57**	76.02	66.13	**88.36**	**90.96**
SSD-RPA300	300 × 300	VGG16	75.22	80.09	78.98	80.57	70.40	66.03	52.09	83.19	88.89
SSD-RPA512	512 × 512	VGG16	**83.89**	**86.35**	**86.05**	**89.17**	80.10	**77.76**	**67.05**	87.88	90.86

The bold value indicates the highest detection accuracy

Table 3
Real time detection results on GDUT-HWD test set. To accelerate the inference, we merge the parameters of batch normalization layers into the convolution operations ahead. "*" means that we have added the 38 × 38 feature maps for detection compared with the original implementations. The speeds are all measured on a single Quadro M2000M 4GB GPU.

Method	Input size	mAP	Blue	White	Yellow	Red	None	Model size	FPS
SqueezeNet-SSD [33]	300 × 300	61.13	68.50	62.48	68.85	59.18	46.63	**13.9 MB**	5.76
YOLOv3-tiny [34]	416 × 416	63.44	71.83	66.75	67.05	58.75	52.85	34.7 MB	**31.30**
MobileNet-SSD* [35]	300 × 300	58.35	65.41	59.02	63.89	57.29	46.14	22.6 MB	3.12
Pelee* [32]	304 × 304	71.68	76.47	74.41	78.43	67.93	61.17	22.5 MB	3.29
Pelee-RPA	304 × 304	**72.34**	**77.35**	**74.64**	**78.48**	**68.58**	**62.67**	36.4 MB	3.22

The bold values indicate the highest performance

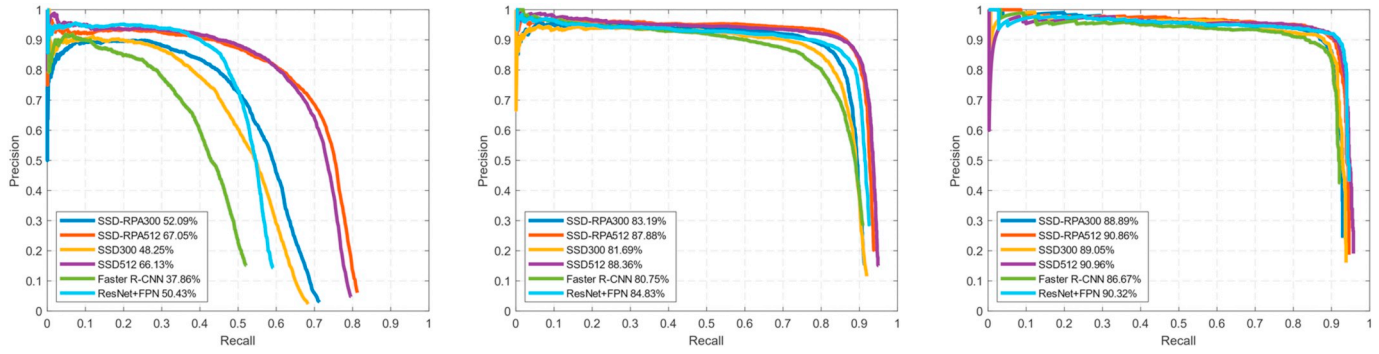


Fig. 3. Simultaneous hardhat wearing detection for small, medium and large instances.

the *Precision* × *Recall* curves for small, medium and large instances. As shown in Fig. 3, it can be seen that, compared with the detections for medium and large objects, the small hardhats detection is quite a challenge due to the vanishing information in top layers. Nonetheless, the proposed RPA can boost performance in the manner of aggregating multi-level features discriminately, which shows an improvement for classes and small objects.

4.3.2. Lightweight detector baselines

To meet the industrial requirements of real-time processing and low memory, we have released the lightweight hardhat detection models. These models achieve a much better trade-off between resources (model size & FPS) and detection accuracy (mAP), which is extremely useful for resource-restricted usages. Among them, our RPA embedded in Pelee [32] (Pelee-RPA) outperforms others in terms of detection accuracy while still maintains competition in speed and memory storage. The results are reported in Table 3.

4.4. Ablation study

In order to understand the effectiveness of our RPA, we conduct experiments to examine how each stage of RPA affects performance. To measure the advantage gained, we progressively remove RPA modules and compare results. For fair comparisons, we follow the common training strategy, which is to train our model on the GDUT-HWD trainval set with the same learning configurations. The trained model is tested on the GDUT-HWD test set. Table 4 shows a decrease in mAP with fewer RPA modules, dropping monotonically from 75.22 to 74.66. The results demonstrate that discriminately backpropagating features from top layers to bottom layers boosts performance effectively with only a small additional computational cost.

Table 4
Effects of various stages of RPA on performance.

RPA modules					mAP
RPA-1	RPA-2	RPA-3	RPA-4	RPA-5	
✗	✗	✗	✗	✓	74.66
✗	✗	✗	✓	✓	74.76
✗	✗	✓	✓	✓	74.85
✗	✓	✓	✓	✓	74.95
✓	✓	✓	✓	✓	**75.22**

The bold value indicates the highest detection accuracy

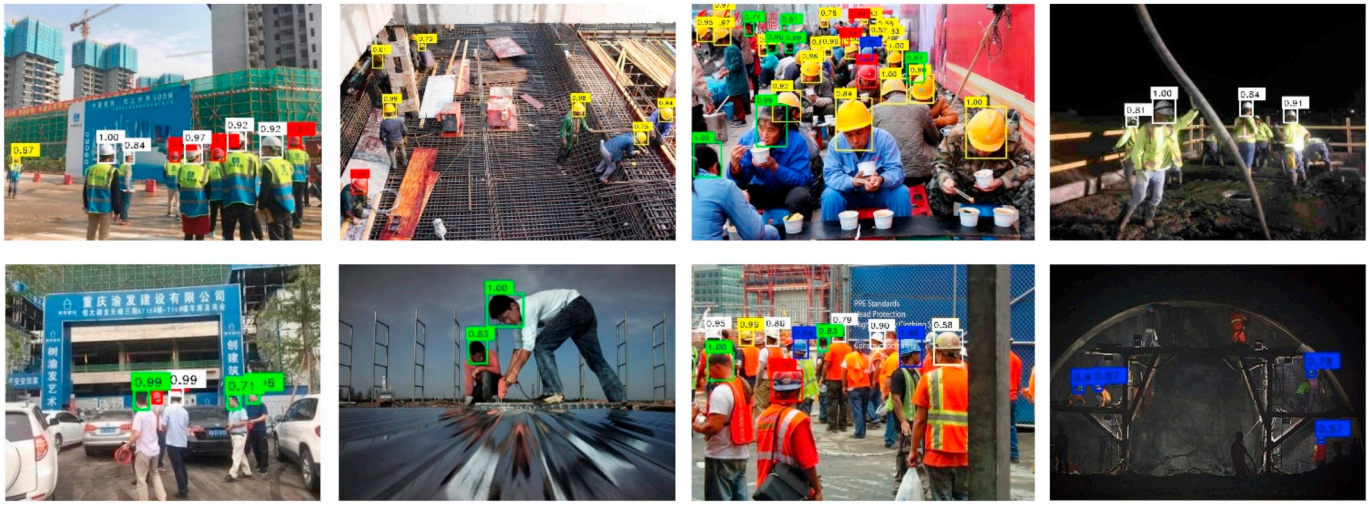


Fig. 5. Detection examples on GDUT-HWD test with SDD-RPA512 model. We show detections with scores higher than 0.5. Each color corresponds to an object category. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4.5. Visualization

In Fig. 4, we show some detection examples on GDUT-HWD with the SSD-RPA512 model. The examples cover a number of factors, including visual ranges, illuminations, individual postures and occlusions. These visualization results indicate the potential abilities of our proposed model that generalizes well to all kinds of construction sites.

5. Discussions

An automatic system for identifying non-hardhats-use provides an effective means of reducing the risk of traumatic brain injury and enhancing construction sites safety. Previous studies have noted the importance of detecting hardhats via computer vision techniques rather than sensor-based methods. However, majorities of studies have adopted multi-stage methods to solve this problem with limitations in adaptation and practical feasibility to construction sites conditions. Meanwhile, in reviewing the literature, no benchmarks have been published in this area and real industrial data are impossible to obtain because of confidentiality constraints. This study set out with the aim of identifying any individuals' hardhat-use by a CNN-based method from on-site scenes. Accordingly, extensive experiments on the proposed benchmark dataset GDUT-HWD demonstrate the effectiveness of the proposed method.

In this study, CNN-based hardhat wearing detection methods are found to be significantly reliable and stable in a wide range of on-site conditions, such as visual range, illumination, individual posture, and occlusions. On the question of identifying small-scale individuals' hardhat-use in far-field images, this study finds out that features aggregation with the proposed reverse progressive attention (RPA) in the SSD framework can boost performance. Evaluation of the GDUT-HWD test set of 1587 images and comparisons with existing prevalent object detection models trained on GDUT-HWD trainval demonstrate its superiority. It may be explained by the fact that increasing context information in feature maps of high resolution substantially improves small-scale objects detection accuracy. By concatenating multi-level features discriminately with RPA, bottom layers introduce more context and spatial information of small-scale objects. Such a method would be much more lightweight and low computational cost compared with concatenating multi-level features directly.

Our study shares similarity with other researches based on deep learning frameworks. For example, Fang et al. [17] trained a Faster R-CNN model to detect non-hardhat-use. However, our study is significant in at least three major respects.

1) The proposed dataset GDUT-HWD is more challenging. First, this requires detecting all the hardhat-use and non-hardhat-use workers and identifying the corresponding hardhat colors while dataset by Fang et al. is only used to detect NHU workers. Hardhat colors can signify different roles on construction sites, which provide an accessible way to enhance construction safety management. Second, the ground truth bounding box annotation for the head region of a worker in our dataset is more reasonable than that for the whole workers in the dataset by Fang et al. Dataset by Fang et al. heavily relies on the bounding box annotation for the whole workers, which inevitably incorporates background pixels that usually occupy more than half of the rectangular area, introducing ambiguities and uncertainties to confuse detection. Instead, the annotation for the head region of a worker makes the detection results less sensitive to occlusion and part of head visible cases, making the detectors more robust.
2) Experimental results show that our models are more effective compared to the Faster R-CNN method adopted by Fang et al. According to Table 2, our SSD-RPA300 exceeds the Faster R-CNN by 9.55% (75.22% vs 65.67%) in mAP and 14.23% (52.09% vs 37.86%) in AP for small instances when both models are trained and tested on the GDUT-HWD. Not only that, the Faster R-CNN model has the worst performance, which may ascribe to the limitation in transforming the single-scale feature maps for detection while others methods make full use of multi-scale feature maps.
3) To assist research in this field, data, trained models and source code are publicly available. To our knowledge, this is the first research to detect both the hardhat-use and non-hardhat-use workers and apply classification to hardhat colors simultaneously in a single model. This can be used as baseline for future research.

In future investigations, it is recommended to extend the GDUT-HWD dataset with ROI segmentation annotations and to perform multi-task learning on an object detection task and a semantic segmentation task. This helps to detect small-scale hardhats and improve generalization performance.

6. Conclusions

Enhancing on-site individuals' safety is an essential requirement for developing intelligent construction sites. To reduce the fatal injury rate for the construction industry, hardhats are indispensable equipment to protect individuals from brain injury during inspection and operation.

This paper proposes a CNN-based method to automatically check

whether individuals on construction sites are wearing hardhats and identify the corresponding colors of hardhats (blue, white, yellow, and red). First, a new benchmark dataset GDUT-HWD has been divided into a training set and a test set to develop and evaluate various CNN-based object detection models for hardhat wearing detection. Second, to better detect small-scale hardhats, the proposed reverse progressive attention (RPA) is integrated into the SSD framework to propagate context information back to bottom layers discriminately. The test results show that our RPA-SSD models could achieve excellent performance with different input sizes under a variety of conditions. This work contributes to existing knowledge of hardhat wearing detection by providing the first publicly available dataset and offering real-time models for on-site monitoring. Currently, there is a need to further improve accuracy for the small-scale hardhats detection and to combine the hardhats detection with object tracking techniques for real-time safety monitoring.

Acknowledgments

This work was in part supported by the National Natural Science Foundation of China (Grant Nos. 61001179 and 61471132) and the Key Project of Industry-University-Research Collaborative Innovation in Guangzhou, China (Nos. 201802020010 and 201802020007).

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